

Shape of horizontally focusing mirror for bending magnet

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Abstract. A method to define the shape of a free form mirror has been described. Such a mirror can be used to focus synchrotron radiation produced by a charged arc beam. A Monte Carlo simulation is used to test the mirror and compare it to standard elliptical mirrors. Some results are shown with varied parameters. The mirror's focusing accuracy seems to be independent of the central angle or length of the arced beam. The focussing accuracy instead seems to be dependent on the natural angular divergence of the synchrotron radiation. Further research is needed to assess the potential use of these mirrors.

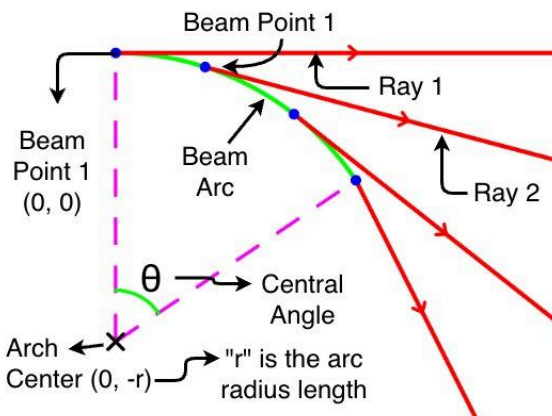
1: Introduction

A method to define the shape of a specific free form mirror is developed. The mirror's purpose would be to accept synchrotron radiation and focus it to a desired focal point. The radiation will be emitted by electrons travelling at relativistic speeds in an arced path accelerated laterally by bending magnets. The reading of the report titled "New focusing mirror system for synchrotron radiation infrared beamlines" [1] up to and including part "2 Optical Layout" is recommended as it was my inspiration and may help better understand this report.

2: Model

A model is created to simplify and communicate methods. The model is a Cartesian plane with x and y dimensions each incrementing in meters. The arced path of the electron (beam arc) due to the bending magnets is coplanar on the Cartesian plane. The starting point of the beam arc (where electrons enter the magnetic field) is the origin of the Cartesian plane. The beam arc is modelled as a curved line turning clockwise starting from the origin. At the origin the tangent of the beam arc is parallel to the x axis. Multiple points are placed on this line spaced equally apart. These points will now be called beam points. Each of these points is modelled to emit one ray with the ray's direction tangent to the curved line at that point. The following parameters will be variable in the model: the number of beam points; the radius of the beam arc; the central angle of the beam arc.

Figure 1: Diagram of beam arc model



The position of the nth beam point is given by:

$$(x, y) = (\sin((n-1)\theta) \times r, \cos((n-1)\theta) \times r - r)$$

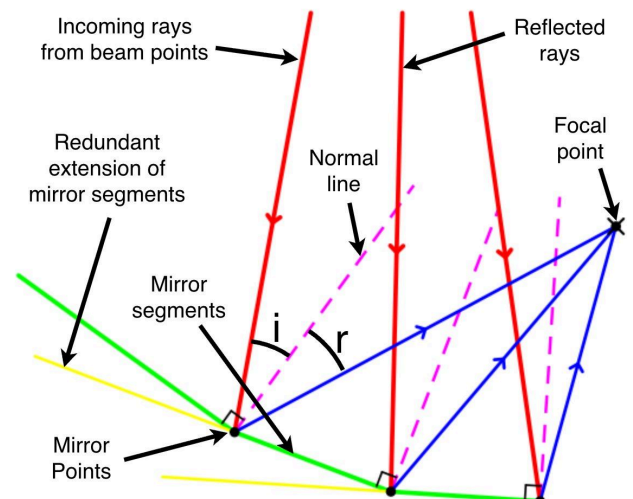
3: Method to define mirror surface shape

The mirror will be modelled as a polyline consisting of points and segments which will be called mirror segments and mirror points. Each mirror segment will connect one point to the next. The first point will start at coordinates (D, 0) where D is a parameter defining the distance of the mirror from the beam arc. This means that the ray emitted by the first beam point [placed at coordinates (0, 0)] will emit a ray directly towards the first mirror point.

Then the following loop is iterated:

1. The current ray is assumed to hit the current mirror point.
2. The direction of the mirror segment is calculated by:
 - 2.1. Finding the angle made by tracing from the beam point, to the mirror point, and then to the focal point.
 - 2.2. Finding the angular bisector of that angle and calling it the normal line.
 - 2.3. The direction of the mirror segment is normal to the normal line.
3. The new mirror segment is then traced out from the current mirror point until it intersects with the path of the next ray.
4. At this intersection, a new mirror point is placed.
5. This loop is repeated with the new mirror point and new ray until it reaches the last ray where the last mirror point is created.

Figure 2: Diagram of mirror generation

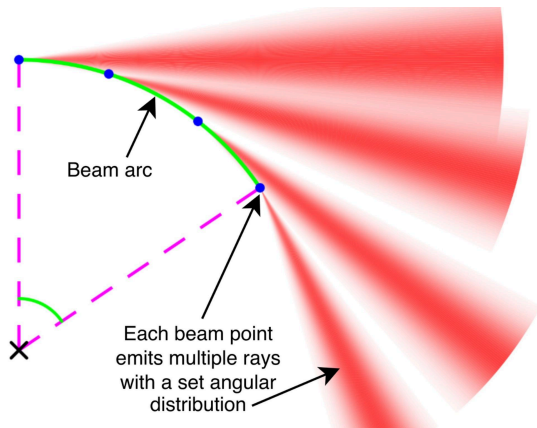


Then a polynomial regression of the desired degree is performed on the coordinates of the mirror points. This new polynomial function will now define the shape and position of the surface of the mirror. The following parameters are variable in this method: the distance (D) of the furthest points of the mirror and the beam arc; the coordinates of the desired focal point; the degree of polynomial to define the mirror surface shape with. A mirror surface shape found by using this method will now be called Arc Focussing Mirror (AFM).

4: Method to test mirror surface shape

an AFM can be tested using a Monte Carlo simulation. A model similar to what is previously described can be used with slight variation. The beam arc can remain the same but the number of beam points (and therefore their spacing) must be changed. This is to make sure the mirror does not only focus rays coming from specific points. Each beam point should emit a number of rays in random angles but with a gaussian or other angular probability spread (natural angular divergence).

Figure n: Rays being emitted in Gaussian distribution



The mirror surface is now modelled as a polynomial function rather than the polyline used before. Each ray is traced until it intersects with the mirror surface curve where they are reflected according to the angle of incidence. If there have been no errors in the defining or testing of the mirror surface, then all the reflected rays should pass very close to the focal point. The average angle of all the reflected rays is calculated. Perpendicular to this angle a screen is placed on the focal point. An intensity graph can be plotted to show the distribution of light rays on the screen. This graph can be used to compare focussing accuracy between different mirror shapes. The graph can also be used to compare AFMs but with different degrees of polynomials. The following parameters are variable in this method: the coordinates of the desired focal point; the number of beampoints; the amount of rays emitted by each beampoint and its angular spread.

5: Computing the methods

A python program is written to carry out the method to define the mirror surface, and test the defined shape. This program can be found on Github [2]. Instructions on how to use the program and details on how the program was made can be found in the README file. The results presented in the next section will be from this program unless mentioned otherwise.

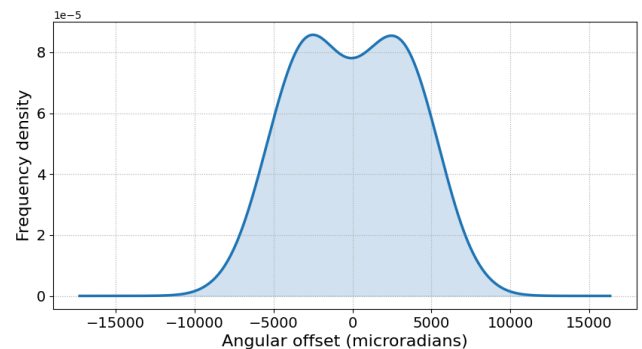
6: Results and analysis

The program mentioned above is used to define and test the mirror surface shape. Some of the parameters are listed below.

- Radius of beam arc: 11.5 meters
- Central angle of beam arc: 8 milliradians
- Distance between beam and mirror (D): 15 meters
- Coordinates of focal point: (15, 15)
- The degree of the polynomial that will describe the shape of the mirror will be variable.

The angular spread of the rays emitted by each beam point is shown below. It was found by using the SHADOW ray-tracer simulation using the OASYS software. The non-default parameters input into the simulation were a beam energy of 1.2 GeV; a beam current of 150 mA and a bending magnetic field of 0.35 T.

Figure 1: Natural angular divergence used for the simulation



A table presenting the results of testing AFMs of different degrees and an elliptical mirror for comparison is presented below. The focal points of the elliptical mirrors are set to (0, 0), which is the start of the beam arc, and (15, 15) which is the desired focal point. The minimal distance required to capture 50% of the rays is used as a benchmark.

Table 1: Results of simulation for 8 mRad beam arc

Shape of mirror	50% collection distance (μm)
1 degree AFM	193755
2 degree AFM	1074.12
3 degree AFM	106.55
4 degree AFM	107.62
5 degree AFM	107.49
6 degree AFM	107.502
Elliptical	263.038

The results show that for these parameters, a 3 degree polynomial is optimal for defining the mirror. This optimal AFM focus gives a 50% collection distance that is 60% less than that produced by the elliptical mirror. The graphs of ray intensity on the screen is shown for both these mirrors.

Figure 2: Ray intensity produced by the AFM

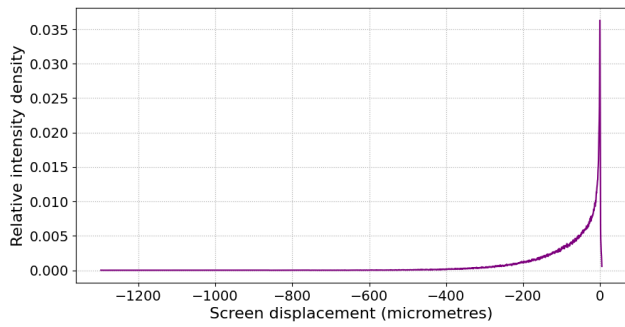
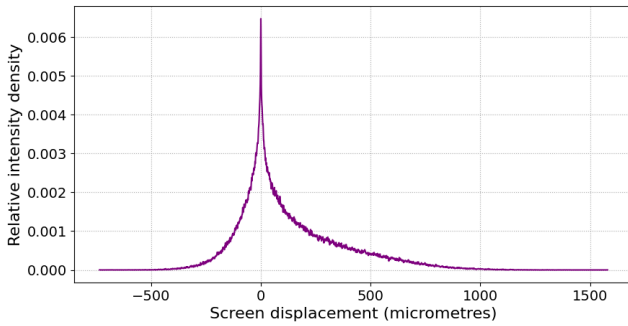


Figure 3: Ray intensity produced by the elliptical mirror



However, the comparison of focussing accuracy changes depending on the central angle (or length) of the beam arc. The central angle of the beam arc is now changed to 2 milliradians and a similar results table is presented below.

Table 2: Results of simulation for 2 mRad beam arc

Shape of mirror	50% collection distance (μm)
1 degree AFM	181642
2 degree AFM	841.991
3 degree AFM	106.11
4 degree AFM	106.711
5 degree AFM	107.969
6 degree AFM	107.856
Elliptical	54.8105

Once again, the AFM defined by a 3 degree polynomial seems to be optimal. However, this time with the reduced central angle, the AFM has a 94% greater 50% collection distance. The central angle is now set to 16 milliradians and another similar results table is presented below.

Table 3: Results of simulation for 16 mRad beam arc

Shape of mirror	50% collection distance (μm)
1 degree AFM	258605
2 degree AFM	1797.12
3 degree AFM	106.11
4 degree AFM	109.785

5 degree AFM	108.981
6 degree AFM	108.668
Elliptical	806.854

The AFM defined by a 3 degree polynomial again seems to have been optimal for a beam arc central angle of 16 milliradians. The 3 degree AFM now requires an 87% smaller distance to collect 50% of the ray compared to the elliptical mirror. An important pattern is seen that the focussing accuracy of the AFM is almost constant regardless of the beam arc length or central angle. The same is not true for an elliptical mirror.

7: Conclusion and discussion

The pattern that an AFM is consistent in focussing accuracy regardless of the central angle and beam arc can be a useful insight. However, to confirm this, more tests need to be done as currently only three configurations verify this. I hypothesize that the focussing accuracy of an AFM is directly related to the natural angular divergence. This is because when defining the mirror, the angular divergence is ignored and only a ray tangent to the beam arc is used. AFMs must be studied and tested more to confirm this. How the central angle and length of the electron beam affects the focusing accuracy of an elliptical mirror is well studied and does not need to be tested. If decreasing the natural angular divergence of the radiation does indeed increase the focussing accuracy of an AFM, then a weaker magnetic field can be used to bend the beam. This is because the weaker field will cause a weaker acceleration to the electrons and therefore cause the longer wavelength radiation to be more collimated. The decreased photon flux caused by decreased magnetic field strength can be compensated for by increasing the length of the beam arc. AFMs perform better than elliptical mirrors in the tests I have performed but the use of other testing methods may also help confirm this. AFMs are likely harder to construct accurately compared to the standard elliptical mirrors. However, in certain cases, this superior focusing accuracy may be required where AFMs may be considered an option. Further experimentation on how the central angle and length of the beam arc affect AFMs' focussing accuracy can reveal potential use cases of AFMs for longer bending magnets. Experimentation on how the natural angular divergence of the synchrotron radiation affects AFMs' focusing accuracy can also help increase radiation intensity at the focal point.

8: Links and References

- [1] S.-J. Chen et al., "New focusing mirror system for synchrotron radiation infrared beamlines," *Opt. Eng.*, vol. 43, no. 12, pp. 3077–3082, Dec. 2004, doi: 10.1117/1.1813438
- [2] R. H. Desai, *AFM_definer_and_tester*, *GitHub*, 2026. [Source code]. Available: https://github.com/chikaskas/AFM_toolkit